ChitChat Business

Feature Documentation & Competitive Analysis

Version 1.1.0 · September 30, 2026

Confidential — Internal Use

1. App Overview

ChitChat Business is a cross-platform mobile application (iOS, Android, Web) built with **React Native / Expo SDK 57** and powered by **Firebase** (Authentication, Firestore, Storage, Cloud Functions). It is a business-focused messaging platform combining real-time chat, voice & video calling, status stories, product catalogues, broadcast messaging, and integrated meeting scheduling.

Platform	iOS, Android, Web (React Native / Expo SDK 57)
Backend	Firebase — Firestore, Auth, Storage, Cloud Functions
Version	1.1.0 (Build 2)
Navigation	React Navigation v7 — Native Stack + Bottom Tabs
Real-time	Firestore onSnapshot listeners
Calling	WebRTC (LiveKit / @livekit/react-native-webrtc)
Auth	Firebase Phone OTP (E.164 format)
Push Notifications	Expo Notifications + Firebase Cloud Messaging

2. Feature Documentation

2.1 Authentication & Onboarding

ChitChat Business uses Firebase Phone Authentication with OTP verification. Every user is tied to a verified phone number in E.164 format.

- Phone number sign-in via SMS OTP (Firebase Auth)
- Automatic session persistence via AsyncStorage across app restarts
- Web sessions expire after 4 hours of inactivity; mobile sessions persist indefinitely
- Biometric lock — Face ID / Touch ID / Android fingerprint (expo-local-authentication)
- Profile creation on first sign-in: display name + profile photo upload to Firebase Storage
- Change phone number flow with OTP re-verification
- Guest/test user bypass (+0000000000) for development
- Offline-resilient: operates in cache-read mode when Firestore is unreachable
- Animated splash screen with spring logo animation and background colour transition

2.2 Messaging & Chat

The Chat screen is the core of the application. It supports one-on-one and group conversations with a rich set of message types and interaction patterns.

Message Types

- Text messages with emoji picker (30 common emoji)
- Image messages — gallery, camera, or file picker
- Voice notes — hold-to-record, swipe-to-cancel, waveform display, auto-retry upload
- File / document messages (PDF, DOCX, etc.)
- Catalog product share — product cards sent inline
- System messages (member joined/left, number changed)

Chat Interactions

- Real-time delivery via Firestore onSnapshot
- Delivery ticks: single (sent) → double grey (delivered) → double blue (read)
- Typing indicators — animated dots while composing
- Reply to specific message with quoted preview (swipe-to-reply)
- React to messages with emoji (long-press)
- Edit sent messages (shown with "edited" label)
- Delete messages — for everyone or just for me
- Forward messages to other chats
- Copy message text to clipboard
- Message info — per-recipient delivery and read timestamps
- Full-text in-chat message search with scroll-to-result highlight
- Date separators between messages crossing calendar day boundaries
- Auto-scroll to latest message on arrival
- Mute individual chats (persisted locally in AsyncStorage)
- Contact presence: online / away / offline with last-seen timestamp
- Block / unblock contacts — bidirectional; blocked messages hidden from recipient

- Full-screen pinch-to-zoom image viewer in chat
- In-chat seller catalog: browse, add to cart, send structured order message
- Cart context: quantity management, order summary

Group Chat

- Create groups with name and multiple participants
- Privacy-aware group creation (respects "Add Me to Groups" setting per contact)
- Group member list with photos and online status
- Past member tracking — shows who left and when
- Group info modal with full member management
- Leave group with confirmation dialog

2.3 Voice & Video Calls

- One-on-one audio calls with native full-screen UI
- One-on-one video calls with camera feed and PiP tap-to-swap
- Incoming call overlay showing caller info with Accept / Decline
- Floating call tile while browsing other screens
- Ringtone playback (plays in silent mode on iOS)
- Audio routing toggle: speaker / earpiece, microphone mute
- Camera toggle during video call
- In-call chat panel — send messages without leaving the call screen
- Call history: incoming, outgoing, missed — with timestamps
- Missed call badge on Calls tab bar icon
- Privacy: "Can Call Me" setting (Everyone / Contacts / Nobody)
- Meeting room support for scheduled multi-party calls

2.4 Status / Stories

- Post image, video, or text-only status updates (24-hour expiry)
- Image statuses display for 7 seconds; video plays to completion (max 30s)
- Progress bar per item with 50ms tick animation; tap to navigate; long-press to pause
- View count and viewer list for own statuses; delete own statuses
- Unviewed status ring on contact avatars in chat list
- Privacy: Who Can See — Everyone / Contacts / Nobody
- Contact-filtering: only shows statuses from saved phone-book contacts
- Image prefetching on list load — media cached before viewer opens
- Cache-aware loading via Image.queryCache — second view shows instantly (no spinner)
- Dark gradient placeholder during initial load instead of black screen
- Video player initialised with URL directly to avoid null-to-replace black frame

2.5 Contacts & Discovery

- Native phone book integration (expo-contacts)
- App users identified by matching phone numbers
- Phone book name resolution overrides Firebase display names

- Online presence indicators on contact rows
- Contact permission request with graceful degradation
- Start new 1-on-1 chat directly from Contacts screen
- Privacy settings respected: profile photo, last seen, status visibility

2.6 Broadcast Lists

- Create named broadcast lists with selected contacts
- Edit list name and recipient set anytime
- Delete broadcast lists
- Compose messages (2000-char limit) with real-time send progress bar
- Each recipient receives the message as a private individual chat
- Last sent timestamp and last message preview per list card

2.7 Product Catalog & Cart

- Products: name, price, description, image (Firebase Storage), collection
- Collections: named categories with emoji for product organisation
- Catalog browsable from chat header and profile modal
- Full-screen catalog sheet with collection tabs and item grid
- Add to cart with per-seller cart and quantity management
- Order summary formatted and sent as a structured message to the seller
- Cart item count badge on chat header
- CatalogBubble: rich product card rendered inline in message list
- CatalogShareSheet: share individual products into any chat

2.8 Calendar & Meetings

- Full calendar view with month/week/day navigation
- Create meetings with title, description, date/time, and participant picker
- Meeting room IDs generated for scheduled conference calls
- In-meeting real-time chat panel (MeetingChatSheet)
- Calendar export to device (expo-print, expo-sharing)
- Native date/time picker (react-native-modal-datetime-picker)

2.9 Privacy & Security

- Last Seen, Profile Photo, Status, Add to Groups, Can Call Me — all configurable: Everyone / Contacts / Nobody
- Read Receipts toggle — turning off also hides others' receipts
- Disappearing Messages — 72-hour auto-delete global toggle
- Blocked Contacts management screen
- Bidirectional block: blocked messages hidden on both sides
- Biometric app lock (Face ID / Touch ID / fingerprint)
- Firebase Auth phone OTP — no passwords stored
- ITSAppUsesNonExemptEncryption: false declared in Info.plist

- Privacy settings synced in Firestore, applied across all sessions in real time

2.10 Notifications

- Push notifications via Expo Notifications + FCM
- Per-device token registered against Firebase UID
- Notification Settings: enable/disable per category, sound, vibration
- In-app toast notifications for foreground alerts
- Unread message badge on Chats tab
- Missed call badge on Calls tab
- Unviewed status dot on Status tab

2.11 Appearance & Theming

- Light and Dark mode (ThemeContext)
- Glassmorphism design: frosted glass cards, gradient pill navigation bar
- Custom font family selection
- Theme persisted across sessions
- Web platform: CSS viewport lock, gradient background override

2.12 Linked Devices

- View and manage devices connected to the account (screen present; multi-device sync in progress)

2.13 Call History

- Full call log: incoming, outgoing, missed with caller name, avatar, type icon, date/time
- Tap a call record to initiate a new call to the same contact
- Call history stored in Firestore for cross-device persistence

3. WhatsApp Business Competitive Analysis

Feature-by-feature comparison between ChitChat Business and WhatsApp Business (App tier). WhatsApp Business API features are noted where relevant.

3.1 Features ChitChat Business Has That WhatsApp Business Does Not

✦ In-app Shopping Cart & Order Flow

ChitChat includes a full cart with quantity management and formatted order messages sent directly to the seller. WhatsApp Business has a catalog but no cart or checkout within the app.

✦ WebRTC Calling (Open Architecture)

ChitChat uses LiveKit/WebRTC — open, configurable, integrable with any TURN/STUN server. WhatsApp's calling infrastructure is proprietary and closed.

✦ Calendar & Meeting Scheduler

Dedicated Calendar tab with meeting creation, participant invites, meeting rooms, and in-meeting chat. WhatsApp Business has no scheduling or calendar feature.

✦ In-Call Chat Panel

Send and receive messages while on a video or audio call without leaving the call screen. WhatsApp does not support in-call messaging.

✦ Multi-party Meeting Rooms

Generates shareable meeting room IDs for conference calls via useMeetingRoom. No equivalent exists in WhatsApp Business.

✦ Full-Text In-Chat Message Search

Real-time message search within a chat with scroll-to-result and animated highlight. WhatsApp Business does not offer in-chat message search on mobile.

✦ Past Group Members Tracking

Group info shows users who have left the group and the date they left. WhatsApp Business does not show past group member history.

✦ Biometric App Lock (all platforms)

Face ID, Touch ID, and fingerprint unlock via expo-local-authentication available from launch. WhatsApp Business added this feature later and inconsistently across platforms.

✦ Custom Theme Engine

Full light/dark mode with glassmorphism, custom font families, and gradient backgrounds — all user-configurable. WhatsApp Business only follows the system-level dark/light mode setting.

✦ Per-Chat Mute (Local Persistence)

Individual chat mute stored in AsyncStorage, surviving restarts without server sync. WhatsApp mute is server-synced and less granular.

✦ Disappearing Messages (72h fixed window)

72-hour auto-delete window — a deliberate middle ground between WhatsApp's 24h and 7-day options.

✦ Web Platform (Standalone App)

ChitChat runs natively in a web browser (React Native Web) as a full standalone app. WhatsApp Business is mobile-only; WhatsApp Web is a companion, not an independent app.

✦ Open / Self-Hosted Backend

Firebase backend is configurable and under developer control. WhatsApp is locked to Meta's infrastructure with no data portability.

✦ Catalog Browsing Inside Chat

View and add products to cart from the seller's catalog without leaving the current chat thread. WhatsApp requires navigating to the business profile to browse the catalog.

3.2 Features WhatsApp Business Has That ChitChat Business Does Not

✗ Business Profile Verification Badge

WhatsApp Business accounts can apply for a green verified badge next to the business name — a trust signal for customers. ChitChat has no verification infrastructure or badge system.

✗ Structured Business Profile Fields

WhatsApp Business profiles include: business category (industry classification), per-day opening hours with schedule, physical address with Google Maps integration, website URL, and email. ChitChat profiles only have a display name and photo.

✗ Automated Greeting Messages

WhatsApp automatically sends a configurable welcome message to any customer who starts a conversation for the first time. ChitChat has no automated greeting system.

✗ Away Messages / Out-of-Office Auto-Reply

Automatically respond to messages received outside configured business hours. ChitChat has no schedule-based auto-reply or away-message feature.

✗ Quick Replies (/ Shortcuts)

Saved message templates triggered by "/" shortcuts — e.g. "/thanks" expands to a full canned response. ChitChat has no quick-reply or shortcut system.

✗ Message Labels / Tags

Colour-coded labels applied to chats or messages (e.g. "New Customer", "Pending Payment", "Order Complete") for conversation organisation. ChitChat has no labelling or tagging system.

✗ Catalog: Public Storefront Link (wa.me)

WhatsApp generates a public storefront URL customers can visit before starting a conversation. ChitChat catalogs are only viewable inside an existing chat — no pre-conversation discovery.

✗ Catalog: Click-to-Chat Links

Shareable catalog links that non-contacts can use to browse products and start a chat without saving the number. ChitChat has no external catalog link.

✗ Sticker Support

Custom sticker packs — static and animated — both first-party and third-party. ChitChat supports emoji reactions but has no sticker feature.

✗ GIF Search & Sending

Built-in GIF search (GIPHY integration) in the media attachment picker. ChitChat does not support GIF search or sending.

✗ Location Sharing (Static & Live)

Share a current location pin or a real-time live location that updates as the user moves. ChitChat has no location sharing capability.

✗ Contact Sharing (vCard)

Share a saved contact as a vCard message. The recipient can save it to their address book in one tap. ChitChat does not have a contact-share message type.

✗ Polls in Group Chats

Create polls with multiple options and real-time vote counts inside group chats. ChitChat has no polling feature.

✗ Starred Messages / Saved Messages

Star individual messages to bookmark them; access all starred messages from a dedicated list. ChitChat has no starring or saved messages feature.

✗ Message Pinning

Pin up to 3 messages at the top of a chat for quick reference. ChitChat has no message pinning.

✗ Communities (Super-Groups)

WhatsApp Communities group multiple related Groups under an umbrella with an announcement channel and cross-group discovery. ChitChat has individual groups but no community structure.

✗ WhatsApp Channels (Follow & Subscribe)

One-way broadcast channels where an admin posts updates and followers receive them read-only, separate from regular chats. ChitChat's broadcast is sender-to-individual-DM, not a subscribable public channel.

✗ In-App Payments (WhatsApp Pay)

Peer-to-peer and business payments via UPI/bank integration in select markets (India, Brazil). ChitChat has a cart/order system but no payment gateway integration.

✗ Two-Step Verification (6-digit PIN)

An optional 6-digit PIN as a second factor on account registration. Prevents unauthorised re-registration of your phone number on another device. ChitChat relies solely on phone OTP with no secondary PIN.

✗ Storage Manager

A dedicated screen showing storage usage broken down by chat, with bulk-clear options per conversation. ChitChat has no storage management UI.

✗ Multi-party Group Calls (up to 32)

WhatsApp supports group voice calls up to 32 participants and group video calls up to 8. ChitChat's WebRTC currently handles 1-on-1 calls (meeting rooms enable multi-party but are not surfaced as a one-tap group call).

✗ Forwarded Message Label

WhatsApp marks forwarded messages with a "Forwarded" label visible to the recipient; frequently forwarded messages get an additional label. ChitChat forwards messages without labels.

✗ Ads & Entry Points (Click to WhatsApp)

Meta ads that drop customers directly into a WhatsApp chat, plus QR codes and short links for customer acquisition. ChitChat has no ad integration or marketing entry-point infrastructure.

✗ Multi-Agent Shared Inbox (API)

Via the WhatsApp Business API, multiple agents can handle the same business number concurrently with assignment and handoff. ChitChat is single-user-per-account.

✗ Message Templates for Notifications (API)

Pre-approved structured templates for transactional messages (order confirmations, shipping, reminders). ChitChat has no template system.

✗ Chatbot / Workflow Automation (API)

Full programmable chatbot support with keyword triggers, conditional flows, and CRM integration. ChitChat has no automation layer.

✗ Analytics & Reporting Dashboard (API)

Message delivery rates, read rates, response time metrics, and campaign analytics in Business Manager. ChitChat has no analytics or reporting.

3.3 Feature Parity Summary Table

■ Present ■ Absent ■ Partial or Platform-Limited

Feature	ChitChat	WhatsApp Business
Phone OTP Authentication	■	■
1-on-1 Text Messaging	■	■
Group Messaging	■	■
Image / File Sharing	■	■
Voice Notes	■	■
Message Reactions	■	■
Reply to Specific Message	■	■
Edit / Delete Messages	■	■
Forward Messages	■	■
Forwarded Label on Message	■	■
Read Receipts (toggle)	■	■
Typing Indicator	■	■
Status / Stories (24h)	■	■
Broadcast Lists	■	■
Product Catalog	■	■
In-app Shopping Cart	■	■
Cart Order Message	■	■
1-on-1 Audio Call	■	■
1-on-1 Video Call	■	■
Multi-party Group Call	■ Meeting Rooms	■ Up to 32
In-Call Chat Panel	■	■
Calendar & Meeting Scheduler	■	■
In-Meeting Chat	■	■
Business Profile (full fields)	■	■
Verified Business Badge	■	■
Greeting / Away Auto-Messages	■	■
Quick Replies (/ shortcuts)	■	■

Feature	ChitChat	WhatsApp Business
Message Labels / Tags	■	■
Stickers	■	■
GIF Search & Sending	■	■
Location Sharing (static + live)	■	■
Contact Sharing (vCard)	■	■
Polls in Groups	■	■
Starred / Saved Messages	■	■
Message Pinning	■	■
Communities	■	■
Channels (1-way subscribe)	■	■
In-app Payments	■	■ Select markets
Two-Step Verification PIN	■	■
Storage Manager	■	■
Biometric App Lock	■	■
Disappearing Messages	■ 72h fixed	■ 24h/7d/90d
Privacy Controls (granular)	■	■
Blocked Contacts	■	■
In-chat Message Search	■	■
Dark Mode / Custom Theme	■ Full custom	■ System only
Web App (standalone)	■	■ Companion only
Catalog: Public Storefront Link	■	■
Catalog: Click-to-Chat Entry Points	■	■
Ads / Click-to-WhatsApp	■	■
Multi-Agent Inbox	■	■ API tier
Message Templates	■	■ API tier
Chatbot / Automation	■	■ API tier
Analytics Dashboard	■	■ API tier
Open / Self-Hosted Backend	■	■

4. Technical Architecture Summary

Layer	Technology	Notes
UI Framework	React Native 0.86.3 / Expo SDK 57	*iOS, Android, Web from one codebase*
Navigation	React Navigation v7 — Native Stack + Bottom Tabs	*Instant tab switching; native slide for stack pushes*
State Management	React Context + custom Hooks	*AuthContext, CallContext, CartContext, ThemeContext, etc.*
Database	Firestore (NoSQL, real-time)	*onSnapshot listeners, offline persistence*
Auth	Firebase Phone Auth	*SMS OTP, E.164, AsyncStorage session cache*
Storage	Firebase Storage	*Photos, voice notes, status media, catalog images*
Calling	WebRTC — LiveKit	*Peer-to-peer audio & video, STUN/TURN*
Push	Expo Notifications + FCM	*Per-device token, background delivery*
Cloud Functions	Firebase Functions (TypeScript)	*Broadcast routing, scheduled cleanup*
Biometrics	expo-local-authentication	*Face ID, Touch ID, Android fingerprint*
Media	expo-video, expo-audio, expo-image-picker	*Video playback, voice recording, media selection*
Build	EAS Build / Xcode	*Version 1.1.0 (Build 2), appVersionSource: remote*

Firestore Collection Structure

Collection Path	Contents
`users/{uid}`	*Profile, privacy settings, catalog items, collections, presence*
`chats/{chatId}`	*Metadata, members, last message, unread counts, pastMembers*
`chats/{chatId}/messages/{msgId}`	*Messages with type, media URLs, readBy, deliveredTo*
`statuses/{statusId}`	*Media URL, expiresAt, viewedBy, visibility settings*
`calls/{callId}`	*Participants, type, timestamps, WebRTC signalling*
`broadcasts/{listId}`	*List metadata, recipients, lastSentAt, lastMessage*
`users/{uid}/blockedUsers/{targetUid}`	*Block relationships (subcollection)*